Midterm project

2025-10-21

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
stock <- read_csv("World-Stock-Prices-Dataset.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 310122 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (4): Brand_Name, Ticker, Industry_Tag, Country
## dbl  (7): Open, High, Low, Close, Volume, Dividends, Stock Splits
## lgl  (1): Capital Gains
## dttm (1): Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(stock)
```

```
## # A tibble: 6 x 13
##   Date                  Open   High    Low  Close   Volume Brand_Name Ticker
##   <dttm>               <dbl>  <dbl>  <dbl>  <dbl>    <dbl> <chr>      <chr>
## 1 2025-07-03 04:00:00   6.63   6.74   6.61   6.64  4209664 peloton    PTON
## 2 2025-07-03 04:00:00 107.   108.   106.   107.     560190 crocs      CROX
## 3 2025-07-03 04:00:00 123.   123.   122.   122.      36600 adidas     ADDYY
## 4 2025-07-03 04:00:00 222.   224.   221.   223.   29295154 amazon     AMZN
## 5 2025-07-03 04:00:00 212.   215.   212.   214.   34697317 apple      AAPL
## 6 2025-07-03 04:00:00  76.3   77.0   75.6   76.4  11545304 nike       NKE
## # i 5 more variables: Industry_Tag <chr>, Country <chr>, Dividends <dbl>,
## #   `Stock Splits` <dbl>, `Capital Gains` <lgl>
```

Including Plots

You can also embed plots, for example:

```
#install.packages("lubridate")
library(lubridate)
stock$Month <- format(stock$Date, "%Y-%m")
```

Wrangling process

```
stock_prices <- subset(stock, Month > 2016-01)
head(stock_prices)
```

```
## # A tibble: 6 x 14
##   Date                  Open   High    Low  Close   Volume Brand_Name Ticker
##   <dttm>               <dbl>  <dbl>  <dbl>  <dbl>    <dbl> <chr>      <chr>
## 1 2025-07-03 04:00:00   6.63   6.74   6.61   6.64  4209664 peloton    PTON
## 2 2025-07-03 04:00:00 107.   108.   106.   107.     560190 crocs      CROX
## 3 2025-07-03 04:00:00 123.   123.   122.   122.      36600 adidas     ADDYY
## 4 2025-07-03 04:00:00 222.   224.   221.   223.   29295154 amazon     AMZN
## 5 2025-07-03 04:00:00 212.   215.   212.   214.   34697317 apple      AAPL
## 6 2025-07-03 04:00:00  76.3   77.0   75.6   76.4  11545304 nike       NKE
## # i 6 more variables: Industry_Tag <chr>, Country <chr>, Dividends <dbl>,
## #   `Stock Splits` <dbl>, `Capital Gains` <lgl>, Month <chr>
```

```
monthly_price <- stock_prices %>%
  group_by(Brand_Name, Month, Industry_Tag) %>%
  summarize(monthly_price = mean(Close,na.rm =T))
```

```
## `summarise()` has grouped output by 'Brand_Name', 'Month'. You can override
## using the `.groups` argument.
```

```
head(monthly_price)
```

```
## # A tibble: 6 x 4
## # Groups:   Brand_Name, Month [6]
##   Brand_Name Month   Industry_Tag  monthly_price
##   <chr>      <chr>   <chr>                 <dbl>
```

```
## 1 3m         2015-01 manufacturing          122.
## 2 3m         2015-02 manufacturing          126.
## 3 3m         2015-03 manufacturing          125.
## 4 3m         2015-04 manufacturing          123.
## 5 3m         2015-05 manufacturing          121.
## 6 3m         2015-06 manufacturing          120.
```

First plot

```r
par(mar=c(10, 4, 4, 2) + 0.1)
profit <- monthly_price %>%
  filter(Month %in% c("2015-01", "2025-07")) %>%
  group_by(Brand_Name, Industry_Tag) %>%
  summarize(Start_Price = round(monthly_price[Month == "2015-01"], 2),
    End_Price = round(monthly_price[Month == "2025-07"], 2),
    Profit_Rate = round(((End_Price / Start_Price) - 1) * 100, 2))
```

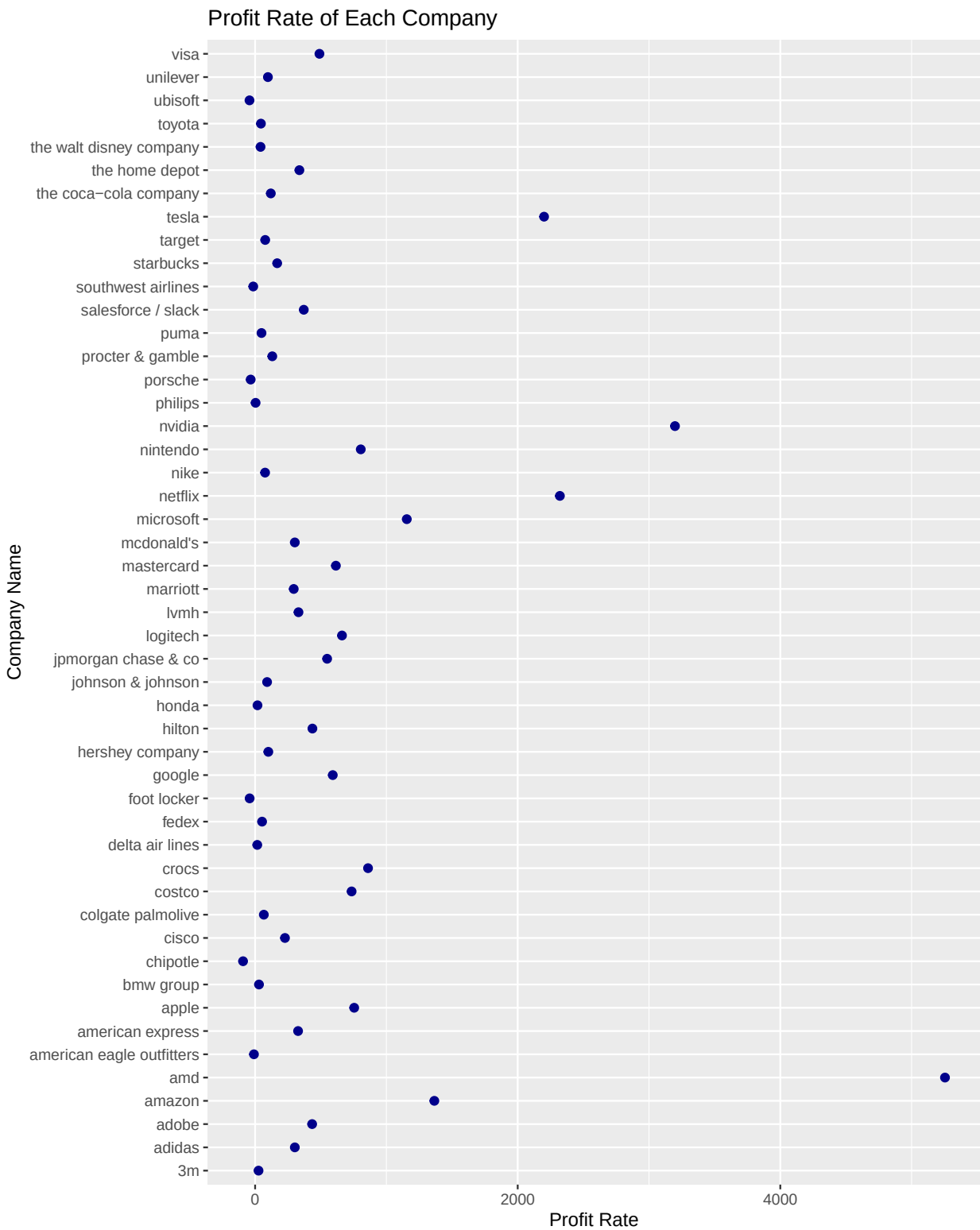
```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `summarise()` has grouped output by 'Brand_Name', 'Industry_Tag'. You can
## override using the `.groups` argument.
```

```r
head(profit)
```

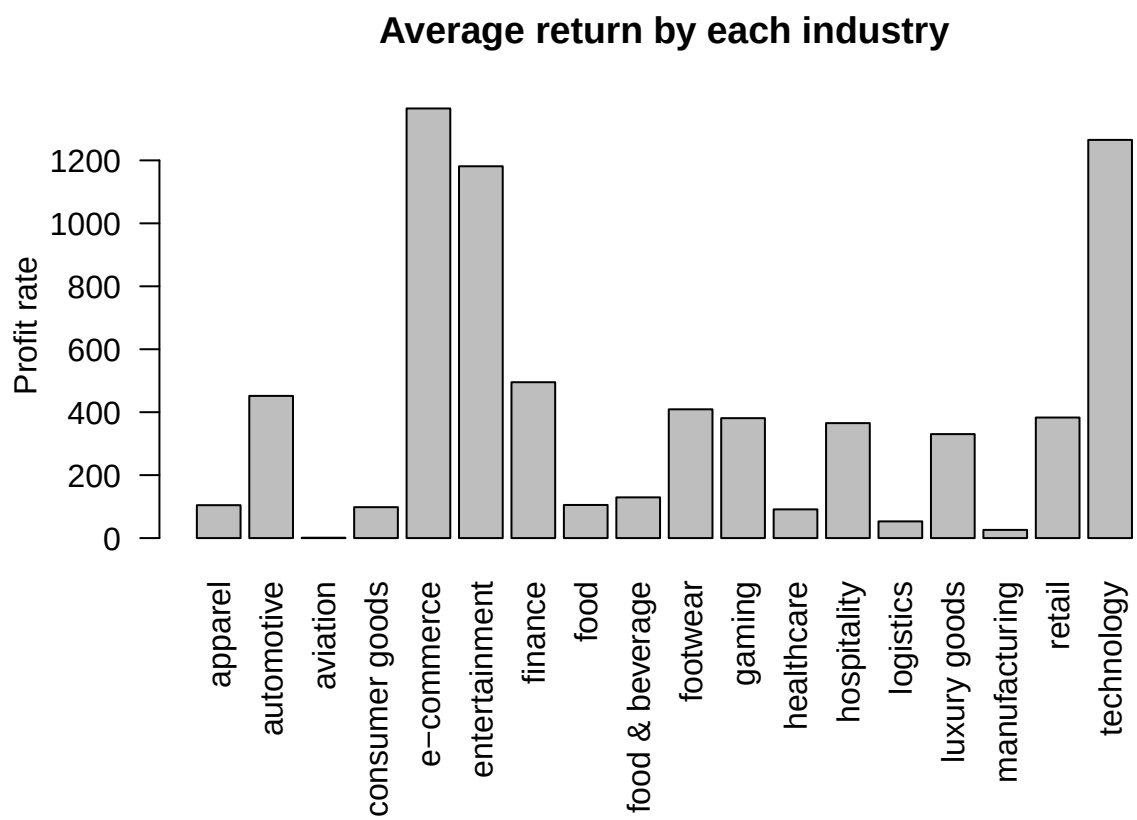
```
## # A tibble: 6 x 5
## # Groups:   Brand_Name, Industry_Tag [6]
##   Brand_Name                Industry_Tag  Start_Price End_Price Profit_Rate
##   <chr>                     <chr>               <dbl>     <dbl>       <dbl>
## 1 3m                        manufacturing      122.       153.        26.0
## 2 adidas                    apparel             30.3      122.       303.
## 3 adobe                     technology          71.6      382.       434.
## 4 amazon                    e-commerce          15.1      222.      1365.
## 5 amd                       technology           2.57     138.      5254.
## 6 american eagle outfitters apparel             11.3       10.3       -8.88
```

```r
ggplot(data = profit, aes(x = Brand_Name, y = Profit_Rate)) +
  geom_point(col = "darkblue", size = 2) + coord_flip() + labs(
    title = "Profit Rate of Each Company", x = "Company Name", y = "Profit Rate")
```



Second plot

```
average_return <- profit %>% group_by(Industry_Tag) %>% summarise(average_return=mean(Profit_Rate, na.r
head(average_return)
```

```
## # A tibble: 6 x 2
##   Industry_Tag   average_return
##   <chr>                   <dbl>
## 1 apparel                 105.
## 2 automotive              452.
## 3 aviation                  1.07
## 4 consumer goods           98.2
## 5 e-commerce             1365.
## 6 entertainment          1181.
```

```
par(mar=c(7, 4, 4, 2) + 0.1)
barplot(average_return$average_return, names.arg = average_return$Industry_Tag,
    xlab = "", ylab = "Profit rate", main = "Average return by each industry",las = 2)
```